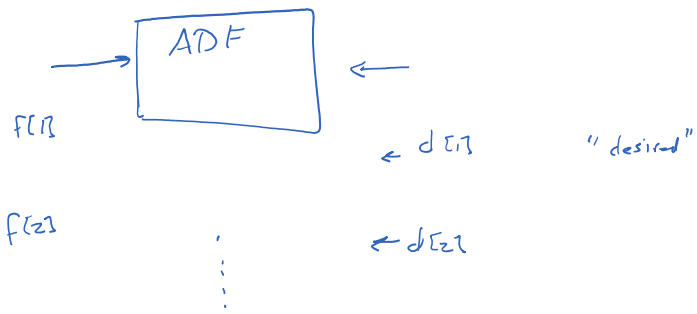
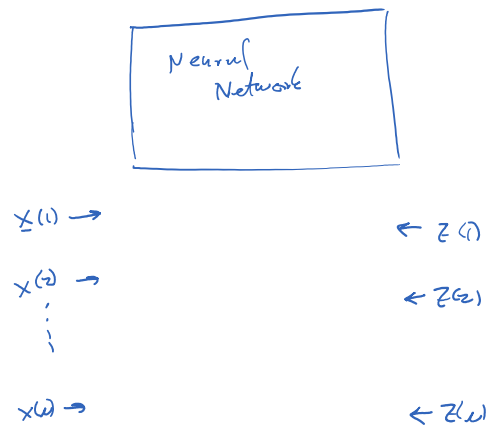
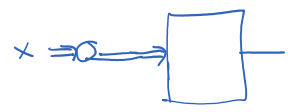
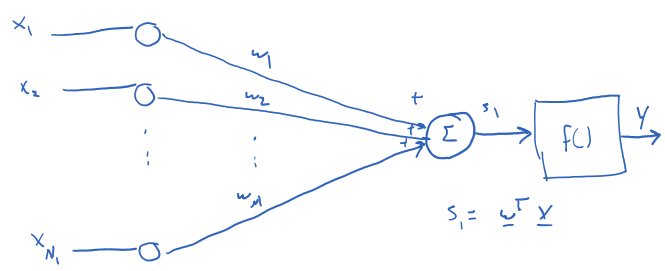




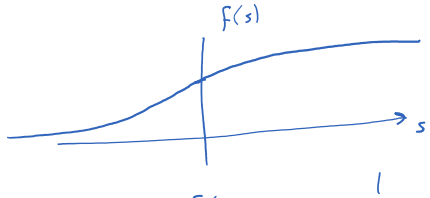
Neuron



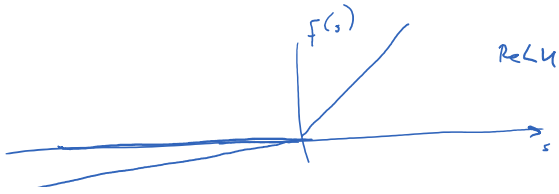


$s_1 = \underline{w}^T \underline{x}$

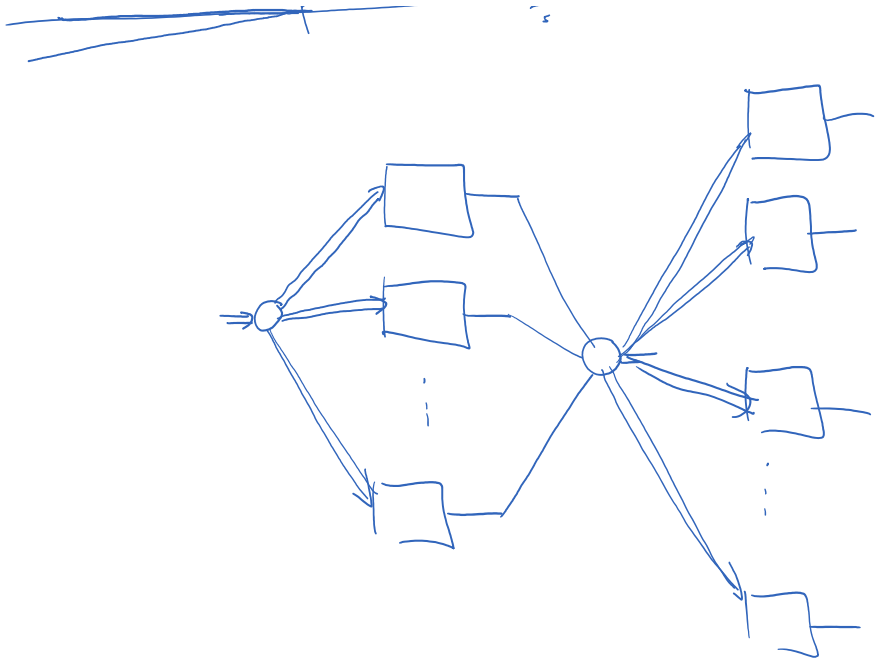
$\underline{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_{N_1} \end{bmatrix}$ $\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}$



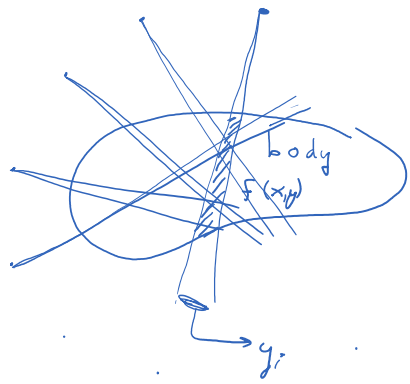
$$f(s) = \frac{1}{1+e^{-s}}$$



ReLU(s)



Natural Pixel Tomography



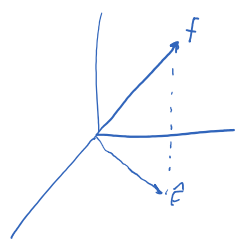
$$\phi_i(x,y) = \begin{cases} 1 & \text{if } (x,y) \text{ is in the beam} \\ 0 & \end{cases}$$

$$y_i = \iint_{\Omega} f(x,y)\, \phi_i(x,y)\, dx\, dy + \underbrace{\nu_i}_{\text{noise}}$$

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}$$

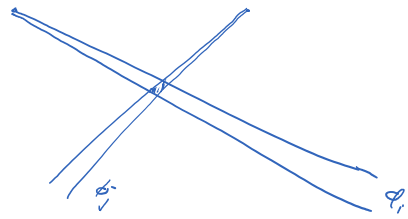
$$\hat{f}(x,y) = \sum_{i=1}^{N} \hat{x}_i\, \phi_i(x,y)$$

How to find $\hat{x}_i$?



$$\begin{bmatrix} \langle \phi_1, \phi_1 \rangle & \langle \phi_1, \phi_2 \rangle & \cdots & \langle \phi_1, \phi_N \rangle \\ \langle \phi_2, \phi_1 \rangle & \langle \phi_2, \phi_2 \rangle & \cdots & \langle \phi_2, \phi_N \rangle \\ \vdots & & & \\ \langle \phi_N, \phi_1 \rangle & \cdots & & \langle \phi_N, \phi_N \rangle \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix} = \begin{bmatrix} \langle f, \phi_1 \rangle \\ \langle f, \phi_2 \rangle \\ \vdots \\ \langle f, \phi_N \rangle \end{bmatrix}$$

$$\langle \phi_i, \phi_j \rangle = \iint \phi_i(x,y)\, \phi_j(x,y)\, dx\, dy$$



$\hat{f}$

$$\min \| f - \hat{f} \|_2^2$$

- $\iint \left(f(x,y) - \hat{f}(x,y) \right)^2 dx\, dy$
- $\langle a(x,y), b(x,y) \rangle = \iint a(x,y)\, b(x,y)\, dx\, dy$
- error $\perp$ data